Sustainable Urban Logistics Optimization

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1. Executive Summary

Table 1. Executive summary of the recommended solution

Metric	Recommended GP solution
Open facilities	L003, L004, L013
Vehicle plan	3 Delivery Truck - Electric vehicles
Demand served	8,500 packages/day; 30/30 zones
Total cost	$2,053.86/day ($0.2416/package)
Cost target	<= $2,100/day; met by $46.14
Priority-adjusted service	1.9963 min; target met
Operating emissions	1.71 kg CO2/day
Environmental compliance	30/30 zones compliant; all LEZ zones served by EVs

The Recommendation

Open three micro-fulfillment centers (L003, L004, and L013) and run a three-vehicle all-electric fleet (one Delivery Truck – Electric per hub). This network serves all 8,500 daily packages across 30 zones at $2,053.86/day with a priority-adjusted delivery time of 2.00 minutes, meeting both of management's targets (cost ≤ $2,100/day; service ≤ 2.0 min), and running entirely on electric vehicles, well within every emission cap. The service figure is a priority-demand-weighted index of facility-to-zone line-haul time used to compare designs, not a door-to-door delivery promise.

Why Not the Cheapest Network

The plan costs $758/day more than the cheapest possible layout (a bare two-site network at $1,296/day), and that premium is deliberate. The cheapest network forces all volume through two sites, leaving outer zones slow (5.1-min service) and the operation exposed to single-point failure. The third site is what delivers the 2-minute promise, the redundancy, and the density buffer for growth. Because roughly 90% of daily cost is fixed facility cost, the decision that creates value is where to open — transport and fleet costs are minor by comparison.

Serving Client C103

C103 can be served on the existing three-site network, no new facility is required, but it is not free, and the right number is not +$209/day. That figure is only the variable transport cost of the extra trips the six-stop cap creates (daily trips rise from 30 to about 108). Sized realistically — round-trip travel plus per-stop service — the workload needs about 10 vehicles, not the three-vehicle baseline (the baseline counts one-way driving only), adding roughly $255/day in fleet capital. Gross all-in is about +$465/day; the contingent EV subsidy (~$227/day, per-trip, unit unconfirmed) brings the net premium to about +$238/day, leaving C103 roughly 9% over the $2,100 target — and driver labor is excluded entirely, so even that is a floor. Budget C103 at the gross transport-plus-fleet cost, size the fleet before committing, and treat the subsidy as upside. The lever that actually lowers C103's cost is delivery density (packages per stop) and a targeted six-stop relaxation in dense zones — not vehicle choice or re-siting.

Where the Plan Is Robust, and Where It Is Fragile

The network is robust where managers instinctively spend, and fragile where they don't. Cost moves less than 4% under demand swings, congestion, or round-trip routing, and the three-vehicle fleet absorbs even severe congestion with zero shortfall — so there is no case for cost or fleet buffers. The real exposures are two: the 2.0-minute service target has zero slack (it slips past target under even mild congestion), and facility capacity becomes binding under a demand surge.

The One Threshold to Watch

The network runs at about 52% of capacity today, but a 2× demand surge is the single threshold that breaks the plan — and it breaks two ways: if the network re-plans, service slips to 2.26 minutes; if it stays committed, L003 overflows to about 186% of capacity. At that point the fix is to bring one pre-identified site, L008, online — holding the existing three, still meeting the service target, at about $2,763/day. This is a contingency to pre-position, not a rebuild of the network.

Bottom Line

Deploy the three-site electric network now, and accept C103 at its gross cost. Then watch two leading indicators — weekly demand against a 1.5× trip-wire (pre-position L008), and priority-zone travel time against free-flow (defend the service SLA through routing and target-setting, not more trucks). The recommendation is financially solid; management's attention belongs on demand capacity and service resilience, not on cost or fleet.

2. Business Problem and Decision Context

2.1 Business context

GreenCity Logistics is a mature last-mile delivery company operating in the dense urban core of Lower Manhattan. Its current delivery network was assembled rapidly during an early expansion phase and has not been systematically optimized around today's operating environment. The company now faces three simultaneous pressures: urban demand growth, city policy that favors low-emission logistics, and competitors that are densifying their fulfillment networks to improve speed and reliability. GreenCity therefore needs to redesign its micro-fulfillment center (MFC) network as a business decision, not only as a transportation problem.

The decision system evaluates a multi-echelon last-mile network in which micro-fulfillment centers serve as the intermediate layer between upstream supply and customer demand zones. The Lower Manhattan service area is compact but operationally complex: demand is spread across 30 zones, priority delivery requirements vary by customer segment, and low-emission zones restrict vehicle choices. Total modeled daily demand is 8,500 packages.

2.2 Decision problem

From 25 candidate MFCs, GreenCity must decide which sites to open. For each of the 30 demand zones, the company must then decide which open facility will serve the zone, which vehicle type will operate the route, and how many trips are required. These decisions are linked. Opening more facilities improves proximity and service resilience, but raises fixed costs. Using electric vehicles supports compliance, but vehicle capacity and work-time assumptions still affect fleet sizing. The model is therefore designed as an integrated decision system connecting facility location, demand allocation, vehicle assignment, cost, service, and environmental feasibility.

2.3 Objectives, scope, and assumptions

GreenCity's redesign balances two objectives — cost and service quality — with emissions held as a hard constraint. Cost is measured as the daily-equivalent system cost, including facility fixed cost, fleet fixed cost, distance operating cost, tolls, and congestion charges. Service quality is measured using priority-adjusted delivery time, so zones with more urgent demand receive greater weight. Emissions are modeled as a hard compliance boundary: every zone must stay within its daily cap, and low-emission zones must be served by electric vehicles.

Table 2. Business objectives and modeling scope

Item	Modeling treatment	Business meaning
Cost	Minimize daily-equivalent total system cost	Protect operating margin and avoid over-building the network
Service	Minimize priority-adjusted average delivery time	Give higher weight to urgent demand segments

Item	Modeling treatment	Business meaning
Emissions	Hard cap and LEZ feasibility constraints	Avoid regulatory violations and support green logistics
Planning horizon	Single representative day	Supports strategic network design using daily demand and daily cost metrics
Demand	Deterministic and zone-level	Focuses the model on network design rather than stochastic dispatch
Assignment	Single-sourcing by zone	Creates clear operating ownership and simplifies implementation

3. Data Foundation for Decision-Making

The analysis uses five core datasets: candidate facility locations, customer demand zones, vehicle fleet options, the completed facility-zone distance matrix, and zone-level environmental constraints. These files are cleaned and integrated into model-ready route-vehicle parameters so the optimizer can evaluate cost, service, emissions, feasibility, and assignment logic simultaneously.

Table 3. Data sources used in the decision system

Dataset	Size / role	Main fields used
Location data	25 candidate MFCs	Capacity, setup cost, monthly lease, utility cost, building type, district, suitability, zoning
Demand data	30 demand zones	Daily demand, priority delivery percent, time windows, delivery difficulty, district
Vehicle data	10 vehicle types	Capacity, fuel type, purchase cost, lifespan years, maintenance cost, CO2/km, operating cost/km, range
Distance matrix	25 x 30 = 750 routes	Distance, travel time, congestion factor, emission factor, toll cost, reliability
Emission constraints	30 zone policies	Emission cap, LEZ flag, congestion charge, sustainability incentive

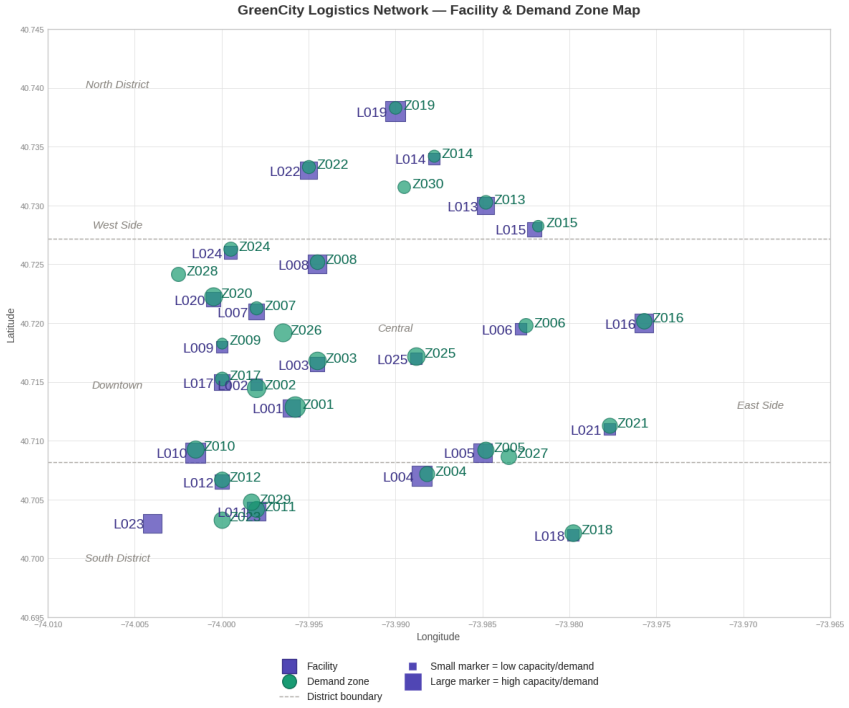


Figure 1. Candidate MFCs and demand zones in the Lower Manhattan network

3.1 Data preparation and integration

Facility, zone, and vehicle identifiers were standardized so records could be joined consistently across datasets. The route table was completed so every candidate facility could be evaluated against every demand zone, producing 750 facility-zone arcs. Facility setup cost, lease, utilities, vehicle purchase cost, and maintenance cost were converted to a consistent daily basis. For each facility-zone-vehicle combination, the model then computed variable transport cost, route emissions, trip requirements, priority-adjusted service burden, EV feasibility, low-emission-zone feasibility, and range feasibility.

The most important derived parameters are daily facility fixed cost, daily vehicle fixed cost, cost per trip, emissions per trip, required trips, weighted service time. These parameters translate business rules into the coefficients used by the LP, network flow, MIP, and multi-objective models. This preprocessing also protects consistency: the same cost and feasibility definitions are used across all model variants.

3.2 Decision-relevant data insights

EDA shows that GreenCity's problem is primarily a facility-siting and service-design problem, not a pure distance-minimization problem. Total candidate capacity is much larger than daily demand, so the question is which subset of facilities should be opened, not whether enough capacity exists somewhere in the system. The compact geography keeps variable distance cost low, while facility fixed cost dominates daily economics. This is why the final recommendation is driven by the number and location of open facilities.

Environmental constraints matter because many zones are low-emission or policy-sensitive, making electric delivery operationally attractive. However, in the final all-electric solution, emissions are far below the caps, so environmental compliance is not binding. The binding managerial issue is service: the cheapest network is too slow, and the recommended network is designed to meet the 2-minute priority-adjusted service target without moving to an extreme high-cost service-focused footprint.

3.3 Data limitations

The distance matrix and route attributes are based on geographic and rule-based calculations, not live traffic observations. Travel times, congestion factors, reliability, and emission multipliers should therefore be interpreted as modeling estimates. The demand data is zone-level rather than stop-level, so the model does not sequence individual deliveries. Labor cost, driver shifts, vehicle charging schedules, and warehouse handling time are not included in the base objective. These limitations do not invalidate the strategic network recommendation, but they should be addressed before operational rollout.

4. Optimization-Based Decision System

GreenCity’s optimization framework is designed as a connected decision system. Each model answers the management question in the decision process. The LP and Network Flow models first diagnose the cost structure and validate the source–facility–zone–sink flow logic. The MIP then converts that network logic into an implementable operating plan by adding facility-opening decisions, zone assignments, vehicle choices, trip counts, fleet sizing, capacity limits, range feasibility, low-emission-zone restrictions, and emission caps. The multi-objective layer uses the MIP structure to evaluate the cost–service trade-off, and Weighted Goal Programming selects the final recommendation based on management’s targets of **$2,100/day** and **2 minutes** of priority-adjusted service. Finally, scenario analysis and the C103 case stress-test the recommended network rather than producing separate, disconnected recommendations.

In short, the model sequence is: **understand cost drivers → find the cheapest feasible network → evaluate the cost–service trade-off → choose the management-preferred solution → test robustness and client realism**.

4.1 Notation and Model Objects

The following notation is used across the core MIP, multi-objective model, Goal Programming model, scenario analysis, and C103 client case. The notation reflects the model-ready sets and parameter dictionaries. LP-specific variables and Network Flow variables are introduced separately in Section 4.2 because those models are diagnostic baselines rather than the final operating model.

Table 4. Core sets and parameters

Symbol	Meaning
I	Set of candidate micro-fulfillment centers
J	Set of demand zones
V	Set of vehicle types
A	Set of facility-zone arcs (j)
A	Set of feasible facility-zone-vehicle triples (v)
F	Set of fleet-eligible facility-vehicle pairs (v)
J^{LEZ}	Set of low-emission zones
V^{E}	Set of electric vehicle types
d_j	Daily package demand in zone j
Cap_i	Daily capacity of facility i
F_i	Daily fixed cost of opening facility i
VFC_v	Daily fixed cost of making one vehicle of type $v$available
c_{ijv}	Variable cost per trip for serving route (j)using vehicle v
e_{ijv}	CO_2 emissions per trip for serving route (j)using vehicle v
M_{ijv}	Number of trips required to fully serve zone $j$from facility $i$using vehicle v
T_{ijv}	Vehicle work time per trip for (v)
B_{ijv}	Priority-adjusted time burden used in the service-quality objective
$ECap_j$	Daily CO_2 emissions cap for zone j
W_v	Available operating minutes per vehicle of type $v$per day
$\overline{N}_{iv}$	Upper bound on the number of vehicles of type $v$that can be assigned to facility i

Table 5. Core decision variables

Variable	Type	Business meaning
y_i	Binary	1 if facility i is opened; 0 otherwise
w_{ijv}	Binary	1 if zone j is served by facility i using vehicle v
r_{ijv}	Nonnegative integer	Number of daily trips on route i-j using vehicle v
N_iv	Nonnegative integer	Number of vehicles of type v assigned to facility i

The overall service metric is:

$$Z_{service} = \frac{\sum_{(i,j,v)\in A_{core}} B_{ijv} w_{ijv}}{\sum_{j\in J} priority_demand_j}$$

A lower service value is better because priority demand is assigned to faster and more reliable facility-zone routes.

4.2 Diagnostic Baselines: LP and Network Flow

The LP and Network Flow models are diagnostic baselines. They are not the final recommendation because they do not make full binary facility and vehicle decisions. Their role is to verify the cost structure and the flow logic before the model is expanded into the MIP.

LP baseline

The LP baseline uses continuous facility-zone flows. It estimates the lowest-cost package allocation when facility usage can be fractional and vehicle choice is summarized through a per-package transport-cost coefficient.

Variable	Domain	Meaning
$Flow_{ij}$	Continuous >= 0	Number of packages shipped from facility i to zone j
$Util_i$	Continuous, 0<= $Util_i$ <= 1	Capacity utilization rate of facility i

$$Z_{LP} = \sum_{i\in I} FixedCost_i Util_i + \sum_{(i,j)\in A} TransCost_{ij} Flow_{ij}$$

$$\sum_{i:(i,j)\in A} Flow_{ij} = Demand_j \quad \forall j \in J$$

$$\sum_{j:(i,j)\in A} Flow_{ij} \leq Cap_i Util_i \quad \forall i \in I$$

$$Flow_{ij} \geq 0 \quad \forall (i,j) \in A$$

$$0 \leq Util_i \leq 1 \quad \forall i \in I$$

$TransCost_{ij}$ is transport-only: distance operating cost, toll cost, and congestion charge. Vehicle fixed cost is introduced later in the MIP when vehicle assignment and fleet sizing become explicit decisions.

Network Flow validation

The Network Flow model expresses the same package movement as a source–facility–zone–sink network. A single arc-flow variable replaces the LP flow and utilization variables. Facility capacity is encoded on the source-to-facility arcs, and zone demand is encoded on the zone-to-sink arcs.

Network object	Meaning
SRC	Source node supplying total daily demand
Facility nodes	Candidate MFCs where capacity can be used
Zone nodes	Demand zones that must be served
SNK	Sink node absorbing served demand
f_{uv}	Continuous package flow on arc (u,v)

$$Z_{NF} = \sum_{(u,v)\in A_{NF}} cost_{uv} f_{uv}$$

$$\Sigma_{v:(u,v)\in A_{NF}} f_{uv} - \Sigma_{w:(w,u)\in A_{NF}} f_{wu} = b_u, \quad \forall u \in N$$

$$b_{SRC} = \sum_{j\in J} d_j, \quad b_{SNK} = -\sum_{j\in J} d_j, \quad b_u = 0 \quad \forall u \in I \cup J$$

$$l_{uv} \le f_{uv} \le u_{uv} \quad \forall (u,v) \in A_{NF}$$

Arc type	Lower bound	Upper bound	Business meaning
SRC -> facility i	0	Cap_i	Facility capacity
facility i -> zone j	0	d_j	Route upper bound
zone j -> SNK	d_j	d_j	Demand must be fully served

The LP and Network Flow baselines use different formulations but validate the same delivery structure. Once this structure is confirmed, the MIP adds the actual yes-or-no operating decisions that GreenCity must make.

4.3 Core MIP Formulation

The MIP is the core operating model. It transforms the diagnostic network into an implementable plan by deciding which facilities to open, which facility-vehicle combination serves each zone, how many trips are needed, and how many vehicles must be deployed. It also enforces capacity, emissions, and fleet-time feasibility.

$$Z_{cost} = \sum_{i\in I} F_i y_i + \sum_{(i,v)\in IV_{fleet}} FC_v^V N_{iv} + \sum_{(i,j,v)\in A_{core}} c_{ijv} r_{ijv}$$

The three terms represent daily facility fixed cost, daily fleet fixed cost, and route-level variable transport cost.

$$\sum_{(i,v):(i,j,v)\in A_{core}} w_{ijv} = 1 \quad \forall j \in J$$

Demand coverage / single-sourcing: every zone is fully assigned to exactly one facility-vehicle combination.

$$r_{ijv} = M_{ijv} w_{ijv} \quad \forall (i,j,v) \in A_{core}$$

Trip linking: trips are scheduled only when the assignment is active, and the number of trips is precomputed from zone demand and vehicle capacity.

$$\sum_{(j,v):(i,j,v)\in A_{core}} d_j w_{ijv} \le Cap_i y_i \quad \forall i \in I$$

$$w_{ijv} \le y_i \quad \forall (i,j,v) \in A_{core}$$

Facility capacity and activation: a closed facility cannot serve demand, and an open facility cannot exceed its daily package capacity.

$$\sum_{(i,v):(i,j,v)\in A_{core}} e_{ijv} r_{ijv} \le ECap_j \quad \forall j \in J$$

Emission compliance: emissions generated by deliveries into each zone must remain within that zone's daily cap. Low-emission-zone and range restrictions are already embedded in the feasible triple set $\mathcal{A}$.

$$\sum_{j:(i,j,v)\in A_{core}} T_{ijv} r_{ijv} \le W_v N_{iv} \quad \forall (i,v) \in IV_{fleet}$$

$$N_{iv} \le \overline{N}_{iv} y_i \quad \forall (i,v) \in IV_{fleet}$$

Fleet feasibility: the total route work time assigned to a facility-vehicle pair cannot exceed the available operating minutes of the vehicles deployed there. Vehicles can be assigned only to open facilities.

$$y_i \in \{0,1\}, \quad w_{ijv} \in \{0,1\}, \quad r_{ijv} \in Z_{\ge 0}, \quad N_{iv} \in Z_{\ge 0}$$

The cost-minimizing MIP provides the cheapest feasible operating network. However, the cheapest feasible network is not automatically the recommended network because GreenCity also values priority service quality. This motivates the multi-objective layer.

4.4 Multi-objective Trade-off and Goal Programming

The final decision uses cost and service as active objectives. Emissions remain hard feasibility constraints in all solutions. The multi-objective stage first maps the cost-service trade-off, and the Goal Programming stage then selects one operating point based on management targets.

$$Z_{cost} = \sum_{i \in I} F_i y_i + \sum_{(i,v) \in IV_{fleet}} FC_v^V N_{iv} + \sum_{(i,j,v) \in A_{core}} c_{ijv} r_{ijv}$$

$$Z_{service} = \frac{\sum_{(i,j,v) \in A_{core}} B_{ijv} w_{ijv}}{\sum_{j \in J} PriorityDemand_j}$$

Epsilon-constraint trade-off model

The epsilon-constraint method generates the Pareto frontier by minimizing cost under different service limits while keeping all MIP feasibility constraints active.

$$Z_{cost} \quad s.t. \, Z_{service} \le \varepsilon \; and \; all \; MIP \; feasibility \; constraints$$

Each point on the frontier represents the lowest daily cost GreenCity can achieve for a given priority-adjusted service limit.

Weighted Goal Programming decision model

Goal Programming converts the frontier into a single recommendation by comparing solutions against management targets.

Target / weight	Value	Business rationale
Cost target g_{cost}	\$2,100/day	Near the practical Pareto elbow and below the intended cost ceiling
Service target $g_{service}$	2.0 minutes	Fast enough to improve on the cost-minimum network without over-expanding facilities
α_{cost}	0.6	Cost control has slightly higher priority
$\alpha_{service}$	0.4	Service remains important for priority clients and competitiveness

$$Z_{cost} + d_{cost}^{-} - d_{cost}^{+} = 2100$$

$$Z_{service} + d_{svc}^{-} - d_{svc}^{+} = 2.0$$

$$0.6 \cdot \frac{d_{cost}^{+}}{2100} + 0.4 \cdot \frac{d_{svc}^{+}}{2.0}$$

$$d^{+}_{cost}, d^{-}_{cost}, d^{+}_{svc}, d^{-}_{svc} \geq 0$$

Only undesirable deviations are penalized. Cost above target and service time above target are undesirable; cost below target or service faster than target is favorable and is not penalized.

Equivalent implementation form: $\sum_{(i,j,v)\in A_{core}} B_{ijv} w_{ijv} + d^{-}_{svc} TPD - d^{+}_{svc} TPD = 2.0\,TPD$

Here TPD is the total priority demand. This equivalent burden-based form avoids dividing by total priority demand inside the solver while preserving the same service target.

The Goal Programming solution becomes the recommended network. Scenario analysis and the C103 client case then evaluate that selected network; they are validation layers rather than separate, disconnected recommendations.

4.5 Nonlinear Congestion Approximation for Stress Testing

The optimization model remains a linear MIP. Congestion is operationally nonlinear in reality, but the base model uses preprocessed travel time, reliability, cost, and emission parameters. Nonlinear congestion is evaluated after the GP recommendation by freezing the selected network and stress-testing service and fleet-time performance under a BPR-style travel-time multiplier.

$$T = T_0\left[1 + \alpha\left(\frac{V}{C}\right)^{\beta}\right], \quad \alpha = 0.15, \quad \beta = 4$$

In the stress test, time-dependent quantities such as service burden and fleet work time scale with the BPR multiplier. Distance-based quantities such as route distance and CO2 emissions do not change. This design keeps the optimization reproducible while still showing whether the committed network is fragile under congestion.

Quantity	Treatment in congestion stress test
Facility openings, assignments, trips, fleet size	Frozen at the GP recommendation
Service burden and fleet work time	Scaled by the BPR travel-time multiplier
Distance operating cost and emissions	Unchanged because they are distance-based
Managerial purpose	Identify whether the selected network misses the service target or needs additional fleet capacity under congestion

We use the Bureau of Public Roads volume-delay function with its standard default parameters ($\alpha = 0.15$, $\beta = 4$). It is applied as a post-hoc nonlinear stress test rather than inside the optimisation, because the core model is a linear MIP in which time enters through preprocessed parameters.

This should be interpreted as a nonlinear approximation used for scenario analysis, not as a full nonlinear optimization model. The distinction matters because the deployable recommendation is generated from a solvable and reproducible linear MIP, while congestion risk is evaluated separately.

5. Recommended Network Design and Results

5.1 Final recommended solution

The final Goal Programming solution opens L003, L004, and L013. The assignment creates clear geographic operating ownership: each selected facility anchors a cluster of demand zones, and every zone is single-sourced to one facility. All 8,500 packages are served and all 30 zones are assigned. In the recommended plan, L003 anchors 12 demand zones, while L004 and L013 each anchor 9 zones.

Table 6. Selected facilities in the final GP recommendation

Facility	District	Capacity	Assigned demand	Utilization	Business role
L003	Downtown	3,800	3,535	93.0%	Dense downtown anchor; high utilization and strong service coverage
L004	East Side	7,000	2,605	37.2%	Large-capacity eastern anchor; protects service and growth optionality
L013	North District	5,500	2,360	42.9%	Northern anchor; reduces travel time to upper demand zones

Table 7. Daily cost breakdown for the recommended solution

Cost component	USD/day	Share
Facility fixed cost	$1,863.38	90.7%
Fleet fixed cost	$109.45	5.3%
Distance operating cost	$4.53	0.2%
Toll cost	$0.00	0.0%
Congestion charges	$76.50	3.7%
Total	$2,053.86	100.0%

The cost breakdown is the key financial insight. Facility fixed cost accounts for about 91% of total daily cost, while distance operating cost is less than 1%. Small routing-distance savings therefore cannot materially change network economics. The strategic decision is facility siting: where to open enough capacity to meet service goals without overbuilding.

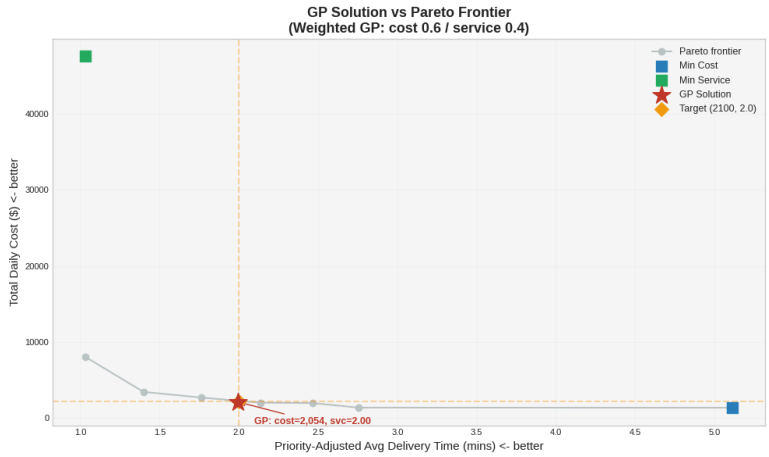


Figure 2. GP solution compared with cost and service benchmark points

5.2 Comparison with benchmark models

Table 8. Model comparison and managerial interpretation

Scenario	Facilities	Daily cost	Service	Emissions	Interpretation
LP / Network Flow baseline	Continuous flow	$724.34	N/A	N/A	Technical lower-complexity benchmark, not deployable
Cost-minimizing MIP	L005, L010	$1,295.73	5.11 min	4.76 kg	Cheapest but too slow and less resilient

Scenario	Facilities	Daily cost	Service	Emissions	Interpretation
Recommended GP solution	L003, L004, L013	$2,053.86	1.9963 min	1.71 kg	Meets cost and service targets with full EV compliance
Service-efficient frontier endpoint	12 facilities	~$8,018	~1.03 min	Compliant	Faster service but high cost; not practical for base rollout
Raw service-only diagnostic	25 facilities	$47,626.55	1.0309 min	2.89 kg	Degenerate service-only point; not a business endpoint

The Pareto frontier shows that service improvement becomes increasingly expensive as GreenCity pushes toward very fast delivery. The recommended GP solution sits near the practical middle of the trade-off: it meets management's 2-minute service goal while staying below the $2,100 cost target and avoiding the extreme cost of a dense service-only network. Note that the LP and Network Flow values are computed on a continuous package-flow relaxation, without binary opening decisions. They are a lower bound on a different cost basis and are not directly comparable to the MIP and GP solutions.

5.3 Environmental and service compliance

The recommended network is environmentally compliant in every zone. Every low-emission zone is served by an electric vehicle, total operating emissions are about 1.71 kg CO2 per day, and all 30 zone-level emission caps have positive slack. Service feasibility is also strong at baseline: the priority-adjusted delivery time is 1.9963 minutes, all zones are assigned, and every package is served. This priority-adjusted delivery time is a priority-demand-weighted index of facility-to-zone line-haul travel time, used to compare network designs; it is not a door-to-door delivery promise. End-to-end delivery time also includes intra-zone travel and per-stop service. However, the service target has almost no buffer, so management should treat travel-time monitoring as an operating priority.

6. Client Case Study: C103 Operational Realism

Operational realism is evaluated using C103, the e-commerce client case. C103 requires at most six stops per trip, electric vehicles in low-emission zones, delivery within two-hour windows, and complete service. Because the dataset is zone-level rather than stop-level, the case is implemented as a re-costing and feasibility audit on the frozen GP network, with an in-model cross-check confirming that the same three facilities remain selected. We assume C103 operates in all zones of the network.

6.1 C103 mathematical representation and assumptions

C103 adds only two new elements to the frozen GP network: a six-stop trip cap and a two-hour delivery-window proxy. The low-emission-zone and complete-service requirements are already satisfied by the GP model and are restated below for verification. C103 specific elements:

Stops: We define stop as a single delivery point inside a zone, e.g. an office building or an apartment-block reception, not the zone itself. A vehicle dispatched to a zone visits several such drop-off points within it. Because the model is zone-level and has no stop-level nodes or sequencing, the six-stop and two-hour results are feasibility proxies on the frozen GP network rather than routing guarantees.

EV subsidy: the unit is inferred. A per-package reading is infeasible (≈ $18,000/day). We adopt a per-trip subsidy applied to electric trips only. It is calculated as a revenue-side offset.

Symbols: i = facility, j = zone, v = vehicle, d_j = daily demand of zone j.

$$n_{\text{stops}}(j) = \left\lceil \frac{d_j}{\text{packages_per_stop}} \right\rceil, \qquad k(j) = \min(6,\ n_{\text{stops}}(j))$$

Estimated stops in zone j, where P_stop is assumed packages per stop. We assume P_stop = 15 packages per stop (based on secondary data estimation; swept over {10, 15, 20} in the cost sensitivity).

$$\text{Trips}_j^{C103} = \left\lceil \frac{n_{\text{stops}}(j)}{6} \right\rceil$$

Six-stop rule: no C103 trip may include more than six different stops.

$$T_{\text{proxy}}(j) = \underbrace{T_{\text{out}}(i,j)}_{\text{facility} \to \text{zone}} + \underbrace{T_{\text{intra}}(j)}_{\text{tour over } k \text{ stops}} + \underbrace{T_{\text{svc}}(j)}_{k \times \text{service}}$$

$$T_{\text{proxy}}(j) \le 120 \text{ min}$$

Two-hour proxy: route time must fit within a 120-minute delivery window. The above three components are defined as follows, evaluated for the worst single trip of at most six stops (return travel is excluded because it does not count against the customer promise):

$$T_{\text{out}}(i,j) = \frac{1.3 \times \text{haversine}(i,j)}{v_{\text{LH}}} \times 60$$

(outbound line-haul; 1.3 = road-circuity factor)

$$v_{\text{LH}} = \text{median}\left\{ \frac{\text{distance_km}}{\text{travel_time_mins}/60} : \text{distance_km} \ge 0.8 \right\} \approx 9.5 \text{ km/h}$$

$$A_j = \pi R_j^2, \qquad L_j = 0.75\sqrt{k\,A_j}, \qquad T_{\text{intra}}(j) = \frac{L_j}{v_{\text{crawl}}} \times 60 \quad [\text{min}] \qquad R_j = \tfrac{1}{2} \min_{k \ne j} \text{haversine}(j,k)$$

(R is taken as half the nearest-neighbour centroid distance,the tour over k uniformly-placed stops)

Line-haul speed v_LH = 9.5 km/h is the median of distance / time over the original 50-route sample. This is deliberately slower than the speed implied by the complete generated distance matrix as using a conservative congested speed makes a two-hour PASS the stronger statement.

Inside a zone the vehicle stops, parks, and re-parks rather than cruising, so the crawl speed must be below the line-haul speed; the 0.5–0.7 range is conventional and we take the 0.6 midpoint, giving v_crawl ≈ 5.7 km/h. Because this term contributes only single-digit minutes, it is held fixed rather than swept.

$$T_{\text{svc}}(j) = k(j) \times s, \qquad s = 7 \text{ min (base)}, \quad s \in \{5, 7, 12\} \text{ (sensitivity)}$$

The 7-minute value breaks down as pure handover plus parking, walk-to-door, and signature time in a dense urban core; it is an engineering estimate, as the dataset provides no per-stop service time. The 30/30 two-hour PASS result holds across the entire {5, 7, 12} range, so it does not depend on the specific value of 7.

Inherited from the GP model:

Electric vehicle requirement in low-emission zones. The LEZ electric-vehicle rule is enforced during data integration by removing any non-electric vehicle on a low-emission-zone route from the feasible set, so the optimizer never sees an infeasible combination.

$$\sum_{(i,v):\,(i,j,v) \in \mathcal{A}} w_{ijv} = 1, \quad \forall\, j$$

Complete service and single-sourcing: every zone is fully served by one assignment.

6.2 C103 results

Table 9. C103 operational realism: GP baseline versus C103 requirements

Metric	GP baseline	C103 case	Delta / interpretation
Facilities opened	L003, L004, L013	L003, L004, L013	No new facility required
Total vehicles	3	10	+7 vehicles under realistic trip-time sizing
Fleet composition	3 electric trucks	10 electric trucks	All-electric network maintained
Trips/day	30	108	+78 trips from six-stop rule

Metric	GP baseline	C103 case	Delta / interpretation
Transport premium	-	$209.28/day	Extra variable transport on added trips
Fleet premium	-	$255.39/day	Capital only; labor excluded
Gross cost	$2,053.86/day	$2,518.53/day	+$464.67/day
EV subsidy	-	-$226.70/day	Contingent; unit undefined in data
Net cost after subsidy	$2,053.86/day	$2,291.83/day	+$237.97/day
Management target	$46.14 under target	$191.83 over target	C103 remains about 9.1% over target
Emissions	1.71 kg	6.15 kg	Still compliant
2-hour feasibility	30/30 zones	30/30 zones	Worst proxy trip about 63 minutes

The C103 case is primarily a workload and pricing issue, not a location-design issue. The same three-facility network remains valid, and all 30 zones pass the two-hour proxy. However, the six-stop rule increases trip count sharply and exposes a fleet-time issue that the baseline model understates because it excludes return travel and per-stop service time. GreenCity should not treat the C103 premium as only a $209 transport surcharge; the realistic gross premium is about $465 per day before driver labor.

The best cost-reduction lever for C103 is delivery density. Increasing packages per stop reduces both trips and fleet requirements. Operationally, GreenCity should pursue building/front-desk consolidation, parcel lockers, scheduled batch drops, and targeted relaxation of the six-stop rule in high-density zones. Vehicle choice helps at the margin, but the cost driver is stops and service time, not distance.

7. Robustness: Sensitivity and Scenario Analysis

The scenario analysis tests whether the recommendation remains valid when management preferences or operating conditions change. The required analysis covers more than two parameters: objective weights, demand level, round-trip routing, congestion severity, and expansion under 2x demand.

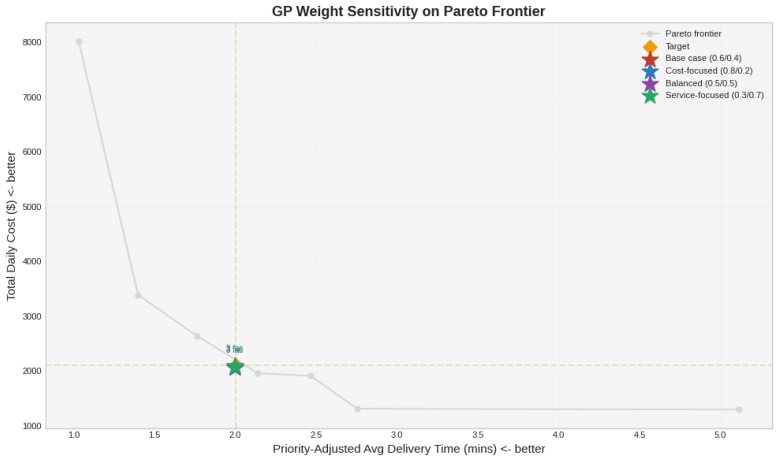


Figure 3. Weight sensitivity: GP solutions remain close to the same facility footprint

Table 10. Goal Programming weight sensitivity

Scenario	Weights / shock	Facilities	Cost	Service	Decision impact
Base GP	0.6 cost / 0.4 service	L003, L004, L013	$2,053.86	1.9963	Recommended plan
Cost-focused	0.8 / 0.2	L003, L004, L013	$2,053.89	1.9963	Facility decision unchanged
Balanced	0.5 / 0.5	L003, L004, L013	$2,084.70	1.9963	Facility decision unchanged

Scenario	Weights / shock	Facilities	Cost	Service	Decision impact
Service-focused	0.3 / 0.7	L003, L004, L013	$2,087.68	1.9963	Facility decision unchanged

The weight sensitivity result is reassuring: reasonable changes in management preferences do not change the recommended facility footprint. This means the three-site recommendation is not an artifact of one arbitrary weight pair.

Table 11. Demand and routing scenario analysis (re-optimized / Lens A: each shock is re-planned from scratch)

Scenario	Facilities selected	Cost	Service	Emissions	Managerial conclusion
Baseline	L003, L004, L013	$2,053.86	1.9963	1.71 kg	Deploy now
Demand downturn (-10%)	L003, L004, L013	$2,081.17	1.9963	1.91 kg	No major redesign required
Peak demand (2x)	L004, L008, L017	$2,058.57	2.2561	3.91 kg	Cost target met but service target missed; re-plan needed
Round-trip routing	L003, L004, L013	$2,070.59	1.9963	3.71 kg	Same footprint remains viable

Note: Table 11 reports re-optimized (Lens A) outcomes — under each shock the model re-plans facilities, fleet, and trips. These differ from the committed/frozen (Lens B/C) figures highlighted in the slide deck, which keep the baseline 3 vehicles / 30 trips. Two cells differ on this basis: round-trip emissions are 3.71 kg re-optimized (41 trips) vs. 3.42 kg frozen (= 2× baseline); and the demand-downturn cost is $2,081.17 re-optimized vs. about −0.2% frozen.

Table 12. Congestion stress test on the frozen GP plan

Congestion scenario	BPR multiplier	Service	Target met?	Fleet feasible?	Interpretation
Free-flow baseline	1.0000	1.9963	Yes	Yes	Meets target with little slack
Mild V/C = 0.9	1.0984	2.1928	No	Yes	Service target breaks before fleet feasibility
Congested V/C = 1.2	1.3110	2.6172	No	Yes	Routing/SLA issue, not a truck-count issue
Severe V/C = 1.6	1.9830	3.9587	No	Yes	Service risk becomes material

The congestion test shows the main fragility of the recommendation. The current fleet remains feasible even under severe congestion, but the service target fails under mild congestion. Management should therefore protect service through routing discipline, time-of-day planning, SLA monitoring, and realistic customer promises rather than by buying trucks as a first response.

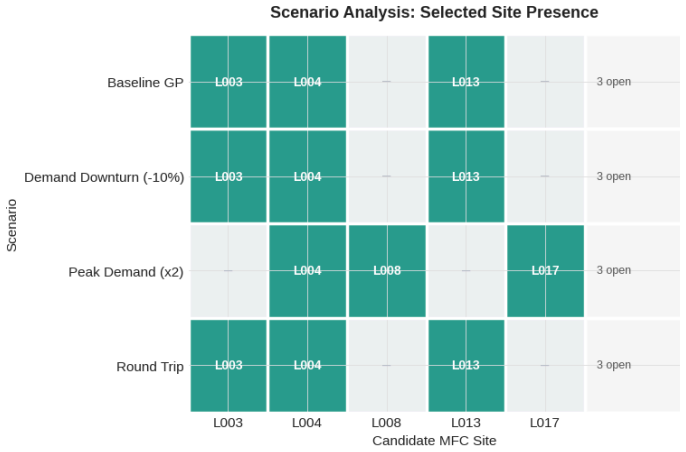


Figure 4. Selected-site changes under demand and routing scenarios

Table 13. Capacity expansion trigger under a 2x demand surge

Expansion case	Opened facilities	Cost	Service	Vehicles	Business implication
Keep-3-add-1 under 2x demand	L003, L004, L008, L013	$2,763.39/day	1.8245 min	4	L008 is the contingency site to pre-position

Demand growth is the structural risk. Under baseline demand, the three-site network is stable. Under 2x demand, capacity and service pressure require an expansion action. The best contingency is not a full rebuild; it is to add L008 while keeping L003, L004, and L013. GreenCity should begin lease-option planning once weekly demand approaches about 1.5x baseline volume, so the fourth site can be activated before a surge forces a rushed redesign.

8. Business Implications and Action Plan

8.1 Managerial insights

Facility siting is the main cost lever. Facility fixed cost represents about 91% of total daily cost. Route distance savings cannot substitute for the right facility footprint.

The third site is a service investment, not waste. The recommended plan costs about $758/day more than the cheapest two-site network, but it improves priority-adjusted service from 5.11 minutes to 1.9963 minutes and creates redundancy.

All-electric delivery is operationally justified. The compact service area and low-emission-zone requirements make electric trucks a practical compliance choice rather than a symbolic sustainability add-on.

Emission caps are not binding; service is. The final solution is comfortably compliant. The operational risk is maintaining the 2-minute service target under congestion.

C103 changes workload, not siting. The same three facilities can serve C103, but the six-stop rule increases trips and fleet needs sharply. Pricing and fleet planning must reflect that burden.

L008 is the named expansion option. At 2x demand, adding L008 to the existing three-site network restores the service target and avoids a full redesign.

8.2 Actionable recommendations

Table 14. Management action plan

Timing	Action	Reason	Metric / trigger
Immediate	Open L003, L004, and L013	Meets cost and service targets with full EV compliance	$2,053.86/day; 1.9963 min service
Immediate	Deploy 3 electric delivery trucks	Sufficient for baseline network and LEZ compliance	1 truck per selected facility
Contracting	Price C103 using gross transport-plus-fleet premium	Six-stop cap creates real workload and fleet requirement	+$464.67/day gross; labor excluded
Operations	Monitor priority-zone travel time daily	Service target has almost no slack under congestion	Trigger routing/SLA action if service >2.0 min
Growth planning	Pre-position L008 as expansion option	2x demand breaks the base network; adding L008 restores service	Start lease option near 1.5x weekly demand
Data governance	Validate subsidy unit and packages-per-stop	These assumptions materially affect C103 economics	Confirm before committing price or fleet

9. Limitations and Extensions

9.1 Limitations

The model is a strategic zone-level network design model, not a full vehicle-routing problem. It decides facility locations, zone assignments, vehicle types, and trip counts, but it does not sequence individual stops. C103 stop counts and two-hour feasibility are therefore proxy-based approximations. A pass means the plan is operationally plausible; it does not replace a stop-level routing schedule.

The cost model excludes driver labor, warehouse labor, charging operations, loading time, IT overhead, and profit margin. This matters most for C103 because the six-stop rule raises daily trips from 30 to about 108 and realistic fleet need from 3 to about 10 vehicles. Labor would increase the true C103 premium beyond the modeled transport-plus-capital estimate.

Demand is deterministic and modeled for one representative day. The model does not include stochastic demand, day-of-week patterns, weather disruptions, or real-time traffic. Route times are generated from the distance matrix and scenario multipliers rather than observed GPS/travel-time data. The final plan should therefore be interpreted as a strategic recommendation that needs operational validation before rollout.

9.2 Extensions

Table 15. Recommended future extensions

Extension	Why it matters
Full VRP with stop sequencing	Converts zone-level assignments into executable driver routes and validates C103 exactly
Labor and shift scheduling	Adds the largest missing operating cost, especially under C103 trip expansion
Time-of-day traffic integration	Improves service-risk estimates and congestion response planning
Stochastic and multi-period demand	Tests whether the network remains robust across weekdays, weekends, seasonality, and promotions
EV charging constraints	Adds charging time, charger availability, and battery management to fleet feasibility
Customer-level client mapping	Separates C103, pharma, grocery, and electronics requirements by actual account geography

10 Appendix

For the full report and additional details, please refer to the Colab notebook.

10.1 Required Output Reports

A1 · Report 1 — Selected Facility Report

Appendix Table A1. Selected Facility Report — recommended GP network.

Facility	District	Building type	Selected	Capacity (pkg)	Assigned demand	Utilization	Setup cost	Monthly operating	Suitability
L003	Downtown	Mixed-Use	YES	3,800	3,535	93.0%	$410,000	$18,200	0.88
L013	North District	Warehouse	YES	5,500	2,360	42.9%	$470,000	$15,600	0.88
L004	East Side	Warehouse	YES	7,000	2,605	37.2%	$520,000	$15,100	0.91

A2 · Report 2 — Demand Allocation Report

Appendix Table A2. Demand Allocation Report — recommended GP network.

Zone	District	Daily demand	Assigned facility	Assigned pkgs	Share	Distance (km)	Travel (min)	Within 2h	Vehicle	Trips	Priority demand
Z001	Downtown	350	L003	350	100.0%	0.42	1	YES	Delivery Truck - Electric	1	147
Z002	Downtown	325	L003	325	100.0%	0.37	1	YES	Delivery Truck - Electric	1	124
Z003	Downtown	310	L003	310	100.0%	0.03	1	YES	Delivery Truck - Electric	1	140
Z007	West Side	260	L003	260	100.0%	0.61	2	YES	Delivery Truck - Electric	1	91
Z009	West Side	240	L003	240	100.0%	0.50	1	YES	Delivery Truck - Electric	1	86
Z010	South District	305	L003	305	100.0%	0.99	3	YES	Delivery Truck - Electric	1	76
Z017	West Side	265	L003	265	100.0%	0.48	1	YES	Delivery Truck - Electric	1	87
Z020	Central	315	L003	315	100.0%	0.82	2	YES	Delivery Truck - Electric	1	132
Z024	West Side	270	L003	270	100.0%	1.17	3	YES	Delivery Truck - Electric	1	92
Z025	Central	310	L003	310	100.0%	0.49	1	YES	Delivery Truck - Electric	1	124
Z026	Central	315	L003	315	100.0%	0.34	1	YES	Delivery Truck - Electric	1	123
Z028	West Side	270	L003	270	100.0%	1.09	3	YES	Delivery Truck - Electric	1	86
Z004	East Side	280	L004	280	100.0%	0.03	1	YES	Delivery Truck - Electric	1	90
Z005	East Side	295	L004	295	100.0%	0.40	1	YES	Delivery Truck - Electric	1	88
Z011	South District	290	L004	290	100.0%	0.86	2	YES	Delivery Truck - Electric	1	64
Z012	South District	285	L004	285	100.0%	0.97	3	YES	Delivery Truck - Electric	1	74
Z018	South District	300	L004	300	100.0%	0.91	3	YES	Delivery Truck - Electric	1	72
Z021	East Side	280	L004	280	100.0%	1.03	3	YES	Delivery Truck - Electric	1	87
Z023	South District	295	L004	295	100.0%	1.05	3	YES	Delivery Truck - Electric	1	68
Z027	East Side	285	L004	285	100.0%	0.46	1	YES	Delivery Truck - Electric	1	83
Z029	South District	295	L004	295	100.0%	0.86	2	YES	Delivery Truck - Electric	1	74
Z006	East Side	270	L013	270	100.0%	1.15	3	YES	Delivery Truck - Electric	1	76
Z008	West Side	275	L013	275	100.0%	0.98	3	YES	Delivery Truck - Electric	1	110
Z013	North District	265	L013	265	100.0%	0.03	1	YES	Delivery Truck - Electric	1	90
Z014	North District	250	L013	250	100.0%	0.53	2	YES	Delivery Truck - Electric	1	95

Zone	District	Daily demand	Assigned facility	Assigned pkgs	Share	Distance (km)	Travel (min)	Within 2h	Vehicle	Trips	Priority demand
Z015	North District	245	L013	245	100.0%	0.32	1	YES	Delivery Truck - Electric	1	78
Z016	East Side	285	L013	285	100.0%	1.33	4	YES	Delivery Truck - Electric	1	83
Z019	North District	255	L013	255	100.0%	1.02	3	YES	Delivery Truck - Electric	1	92
Z022	North District	260	L013	260	100.0%	0.93	3	YES	Delivery Truck - Electric	1	91
Z030	North District	255	L013	255	100.0%	0.43	1	YES	Delivery Truck - Electric	1	94

A3 · Report 3 — Vehicle Assignment Report

Appendix Table A3. Vehicle Assignment Report — recommended GP network.

Vehicle type	Fuel	Vehicles	Trips	Packages served	Total distance (km)	Operating cost	Emissions (kg)	Main districts
Delivery Truck - Electric	Electric	3	30	8,500	20.60	$4.53	1.71	Central, Downtown, East Side, North District, South District, West Side

A4 · Report 4 — Facility-Zone Service Matrix

Appendix Table A4. Facility-Zone Service Matrix — recommended GP network.

Facility	Home district	Zones served	Packages served	Districts covered (packages)
L003	Downtown	12	3,535	Central (940), Downtown (985), South District (305), West Side (1,305)
L004	East Side	9	2,605	East Side (1,140), South District (1,465)
L013	North District	9	2,360	East Side (555), North District (1,530), West Side (275)
TOTAL		**30**	**8,500**	

A5 · Report 5 — Environmental Compliance Report

Appendix Table A5. Environmental Compliance Report — recommended GP network.

Zone	District	Emission cap	Used emissions	Slack	Binding	LEZ	EV used	Congestion charge	Compliant
Z001	Downtown	120	0.034	119.97	unused allowance	Yes	Yes	5.00	OK
Z002	Downtown	120	0.030	119.97	unused allowance	Yes	Yes	5.00	OK
Z003	Downtown	120	0.002	120.00	unused allowance	Yes	Yes	5.00	OK
Z020	Central	130	0.066	129.93	unused allowance	Yes	Yes	4.50	OK
Z025	Central	130	0.039	129.96	unused allowance	Yes	Yes	4.50	OK
Z026	Central	130	0.027	129.97	unused allowance	Yes	Yes	4.50	OK
Z028	West Side	150	0.096	149.90	unused allowance	Yes	Yes	3.50	OK
Z024	West Side	150	0.103	149.90	unused allowance	Yes	Yes	3.50	OK
Z008	West Side	150	0.078	149.92	unused allowance	Yes	Yes	3.50	OK
Z007	West Side	150	0.049	149.95	unused allowance	Yes	Yes	3.50	OK
Z017	West Side	150	0.038	149.96	unused allowance	Yes	Yes	3.50	OK

Zone	District	Emission cap	Used emissions	Slack	Binding	LEZ	EV used	Congestion charge	Compliant
Z009	West Side	150	0.040	149.96	unused allowance	Yes	Yes	3.50	OK
Z019	North District	170	0.090	169.91	unused allowance	Yes	Yes	2.00	OK
Z022	North District	170	0.074	169.93	unused allowance	Yes	Yes	2.00	OK
Z014	North District	170	0.042	169.96	unused allowance	Yes	Yes	2.00	OK
Z015	North District	170	0.026	169.97	unused allowance	Yes	Yes	2.00	OK
Z030	North District	170	0.034	169.97	unused allowance	Yes	Yes	2.00	OK
Z013	North District	170	0.002	170.00	unused allowance	Yes	Yes	2.00	OK
Z016	East Side	180	0.117	179.88	unused allowance	Yes	Yes	2.50	OK
Z006	East Side	180	0.101	179.90	unused allowance	Yes	Yes	2.50	OK
Z021	East Side	180	0.091	179.91	unused allowance	Yes	Yes	2.50	OK
Z027	East Side	180	0.037	179.96	unused allowance	Yes	Yes	2.50	OK
Z005	East Side	180	0.032	179.97	unused allowance	Yes	Yes	2.50	OK
Z004	East Side	180	0.002	180.00	unused allowance	Yes	Yes	2.50	OK
Z023	South District	200	0.092	199.91	unused allowance	No	Yes	0.00	OK
Z010	South District	200	0.079	199.92	unused allowance	No	Yes	0.00	OK
Z012	South District	200	0.078	199.92	unused allowance	No	Yes	0.00	OK
Z011	South District	200	0.069	199.93	unused allowance	No	Yes	0.00	OK
Z029	South District	200	0.069	199.93	unused allowance	No	Yes	0.00	OK
Z018	South District	200	0.073	199.93	unused allowance	No	Yes	0.00	OK

A6 · Report 7 — District-Level Performance Report

Appendix Table A6. District-Level Performance Report — recommended GP network.

District	Daily demand	Facilities serving	Avg travel time (min)	Variable cost	Emissions (kg)	Cost per pkg (var)
Central	940	L003	1.30	$13.86	0.13	$0.0147
Downtown	985	L003	1.00	$15.18	0.07	$0.0154
East Side	1,695	L004, L013	2.20	$15.97	0.38	$0.0094
North District	1,530	L013	1.80	$12.72	0.27	$0.0083
South District	1,770	L003, L004	2.70	$1.24	0.46	$0.0007
West Side	1,580	L003, L013	2.20	$22.06	0.40	$0.0140

A7 · Report 8 — Multi-Objective Trade-Off Report

Appendix Table A7-i. Pareto frontier (ε-constraint method): cost vs. priority-adjusted service.

Epsilon	Cost ($)	Service (min)	Emissions (kg)	Facilities	Vehicles
1.0309	8,018.04	1.0309	0.7383	12	12
1.4022	3,387.75	1.3986	1.2188	5	5
1.7734	2,641.20	1.7616	1.4206	4	4
2.1446	1,960.12	2.1379	1.7654	3	3
2.5158	1,911.32	2.4658	1.9609	3	3
2.8871	1,311.18	2.7538	2.2878	2	2
5.1144	1,295.73	5.1144	4.7556	2	2

Appendix Table A7-ii. Goal-Programming weight sensitivity (cost / service weights).

Scenario	alpha_cost	alpha_service	Cost ($)	Service (min)	Emissions (kg)	Facilities	d_cost+	d_svc+	GP obj
Base case (0.6/0.4)	0.60	0.40	2,053.86	1.9963	1.7107	3	0.00	0.00	0.00
Cost-focused (0.8/0.2)	0.80	0.20	2,053.89	1.9963	1.7302	3	0.00	0.00	0.00
Balanced (0.5/0.5)	0.50	0.50	2,084.70	1.9963	2.0617	3	0.00	0.00	0.00
Service-focused (0.3/0.7)	0.30	0.70	2,087.68	1.9963	1.8149	3	0.00	0.00	0.00

A8 · Report 9 — Scenario & Sensitivity Analysis Report

Appendix Table A8. Scenario & Sensitivity Analysis Report — recommended GP network.

Scenario	Parameter changed	Base → New	Facilities	Cost impact	Service impact	Recommendation changes
Weight 0.8/0.2 (cost-focused)	alpha_cost/alpha_service	0.6/0.4 -> 0.8/0.2	L003,L004,L013	~same	~2.0 min	No
Weight 0.5/0.5 (balanced)	alpha_cost/alpha_service	0.6/0.4 -> 0.5/0.5	L003,L004,L013	~same	~2.0 min	No
Weight 0.3/0.7 (service-focused)	alpha_cost/alpha_service	0.6/0.4 -> 0.3/0.7	L003,L004,L013	~same	~2.0 min	No
Demand downturn	demand factor	1.0x -> 0.9x	L003,L004,L013	2,054 frozen (flat); 2,081 re-opt	2.00 min (met)	No
Peak demand (re-opt)	demand factor	1.0x -> 2.0x	L004,L008,L017	2,054 -> 2,059	2.00 -> 2.26 (breaks)	Yes — restructures
Congestion severe (frozen, BPR)	V/C saturation	free-flow -> V/C 1.6	L003,L004,L013	<+2%	2.00 -> ~3.96 (breaks)	No (routing/SLA, not siting)
Round trip (frozen)	distance basis	one-way -> round-trip	L003,L004,L013	+~0.3%	2.00 min (met)	No (emissions +83%, under cap)
C103 six-stop cap (frozen network)	stops-per-trip <= 6	uncapped -> 6 stops	L003,L004,L013	+465 gross (trans 209 + fleet 255); +238 net of EV subsidy	2.00 min (met)	No siting; fleet 3->~10, ~9% over target
Packages/stop (C103 cost)	packages_per_stop	15 -> {10,20}	L003,L004,L013	transport 158-350/day; fleet ~9-13 veh	n/a	No siting; density is the cost lever

10.2 References: External Benchmarks for Scenario Analysis

To ensure practical robustness, the parameter boundaries for demand shocks and travel-time congestion used throughout the scenario analysis were calibrated against empirical industry benchmarks rather than arbitrary estimates:

- Demand Factors (0.90× & 2.00×)
  - 0.90× Downturn: Reflects the early-2026 consumer spending slowdown and retail stagnation observed in late Q4 2025 (Source data: AOL Retail Sales Report and FinancialContent MarketMinute).
  - 2.00× Peak Surge: Simulates a worst-case promotional single-day peak (e.g., Black Friday), where industry operators typically scale capacity by 2–3× to handle volume spikes (Source data: FreightAmigo E-commerce Guide and FreightWaves Peak Season Analysis).
- Travel Time Multipliers (+30% & ×2)
  - +30% Peak Congestion (V/C = 1.2): Aligns with urban peak-hour delays where standard travel times inflate by approximately 30% compared to free-flow conditions (Source data: TomTom Traffic Index).
  - ×2 Extreme Weather (V/C = 1.6): Functions as a conservative upper-bound stress test combining severe rush-hour traffic with major weather disruptions like heavy rain or snow (Source data: National Institutes of Health (NIH) Transit Efficiency Study).